FITME GYM OPERATING SYSTEM

Next-Generation Smart Gym Management, Biometrics & Payroll Solution

Prepared For: Gym Owners, Directors & Fitness Center Stakeholders | **Date:** October 2026

1. Executive Summary & The 5 Costly Leaks in Traditional Gyms

Traditional fitness clubs in Sri Lanka and South Asia lose an estimated **15% to 30% of gross revenue annually** due to unauthorized access sharing, front-desk cash shrinkage, untracked supplement sales, and manual payroll errors. **FitMe** is an enterprise gym operating system engineered to close every financial leak through automated biometric gates, rigid POS cash drawer controls, real-time member tracking, and Sri Lankan legal payroll (EPF/ETF) compliance.

Traditional Gym Vulnerability	The FitMe Automated Solution	Measurable Financial Impact
Ghost & Buddy Entries: Members share RFID cards or keypad PINs with friends or access facilities after expiry.	**Zero-Touch Face ID / Optical Turnstile:** Gate opens strictly for verified faces/fingerprints with valid, paid active subscriptions.	**+15% to 25% Revenue Recovery: Forces expired and non-members to pay on the spot.**
Front Desk Cash Shrinkage: Cash collected for drop-ins or drinks is pocketed without receipt generation.	**Enforced EOD Cash Drawer Audits:** Opening float, closing declared cash, and automated variance alerts per cashier shift.	**Zero Cash Discrepancies: Unaccounted revenue leaks eliminated completely.**
Supplement & Drink Thefts: Protein scoops, energy drinks, and bars missing without record.	**Touchscreen POS Cycle Audits:** Barcode-ready catalog with automatic discrepancy flags and cost vs. retail margin tracking.	**Protects Retail Margins: Realizes full 40-50% profit margin on gym store sales.**
Payroll & EPF/ETF Disputes: Inaccurate overtime, dispute over trainer commissions, and compliance fines.	**Sri Lankan Statutory Payroll Engine:** Automated 8% EPF, 12% EPF, 3% ETF calculations, PT commissions & thermal advance vouchers.	**100% Legal Compliance: Eliminates labor penalties & saves 25+ hours of HR work monthly.**
Member Churn & Inactivity: Lost contact with members leads to 60%+ non-renewal rates.	**Automated Self-Renewal & Habit Engine:** Sri Lankan nutrition tracker (200+ local meals), habit streaks & WhatsApp/Email notices.	**Higher Lifetime Value (LTV): Boosts member renewal retention by up to 35%.**

2. Deep Technical & Architectural Feature Audit

FitMe has been battle-tested in live gym environments. Below is the architectural breakdown of the 7 core sub-systems:

A. Biometric Access Control & Turnstile Synchronization (`zkbio_integration`)

- **SpeedFace AI Facial Recognition:** Pushes member portrait photos directly to ZKTeco SpeedFace-V5L hardware without keypad PIN risks.
- **High-Speed Optical Turnstiles:** Signals barrier gate relays in <300ms upon successful face or 3-tap fingerprint verification.
- **Real-time Cloud Bridge Daemon (`scripts/zkbio_bridge_daemon.py`):** Runs on Gym PC, constantly monitoring gate punches and syncing to cloud.
- **Offline SQLite Resilient Buffer:** If gym internet drops, punches are queued in local SQLite storage and automatically synchronized upon reconnect.
- **Automated Access Enforcement:** Instantly denies turnstile entry to expired accounts, overdue memberships, or unauthorized off-peak visits.

B. Front Desk Reception & Membership Operations (`membership` + `core`)

- **3-Second Member Lookup:** Instant search across Name, Mobile Phone, NIC / Passport Number, or Member ID.
- **Flexible Subscription Tiers:** Monthly, Quarterly, Bi-Annual, and Annual plans with automated pro-rating and join-date recording.
- **Buddy & Couple Linked Plans:** Couples, training partners, or corporate groups linked to a primary payer with shared or independent benefits.
- **Digital Assessments Workflow:** Electronic intake questionnaires for Workout Readiness & Nutrition, submitted to coaches for digital review.
- **Instant ESC/POS Thermal Invoices:** 80mm thermal receipts with QR-code verification, tax breakdown, and automated PDF email delivery.

C. Retail POS, Bar & Supplement Inventory (`pos`)

- **Touchscreen Cashier Register:** Built for quick sale of pre-workouts, protein scoops, bottled drinks, energy bars, and gym apparel.
- **Omni-Channel Payment Support:** Cash, Card POS terminal, Bank QR transfer, Member Credit Tab, and **Koko Pay** (Buy Now, Pay Later).
- **Cash Drawer Audit Reconciliation:** Tracks opening cash float, register cash ins/outs, closing declared cash, and flags cashier shortages.
- **Stock Discrepancy Audits:** Physical inventory cycle-counting tool with shrinkage flags, restocking alerts, and retail margin analytics.

D. Sri Lankan Statutory Payroll & HR Engine (`gym_operations_payroll`)

- **100% Sri Lankan Labor Compliance:** Automated employee EPF (8%), employer EPF (12%), and employer ETF (3%) statutory calculations.
- **Shift Roster Engine:** Shift templates for Morning (05:00-12:00), Evening (15:00-21:00), Full Split Shifts, and Scheduled Rest Days.
- **Sudden Absence Detection:** Automatically alerts management if a scheduled trainer fails to clock in within 30 minutes of shift start.
- **Salary Advances & Thermal Vouchers:** Front-desk issuance of salary advances with printable thermal voucher signature slips, auto-deducted from final pay.
- **Personal Trainer (PT) Commission Tracking:** Tracks billable personal training sessions per coach and computes accurate revenue splits.
- **80mm Thermal Salary Slips:** One-click printout of itemized monthly pay slips with gross, EPF/ETF, advances, and net salary.

E. Financial Accounting & Daily End-of-Day (EOD) Reports (`core` + `reports`)

- **Automated EOD Daily Financial Close:** Single-sheet breakdown of daily membership revenue, POS retail sales, paid-out expenses, and net cash balance.
- **Club Expense Tracking & Petty Cash Cam:** Front desk can snap photos of grocery/utility/maintenance receipts directly into the digital expense ledger.
- **Accountant Ready CSV Exports:** Instant CSV export for External Auditors (Financial Summary, Member Roster, Payroll Ledger, Expense Log).

F. Sri Lankan Local Nutrition & Habit Engine (`nutrition_lk` + `habit`)

- **200+ Sri Lankan Food Database:** Curated local foods with exact macronutrients (e.g. Red Rice & Fish Ambul Thiyal, Dhal Curry, Kottu, Hoppers).
- **Daily Fuel Targets & Hydration:** Coach-assigned target calories, protein grams, carb grams, and daily water tracking.
- **Workout Streaks & Habit Badges:** Gamifies member attendance consistency to increase club loyalty and community engagement.

G. Multi-Role Portals & Modern Digital Interface (`fitme-ui` + `dashboards`)

- **5 Dedicated Role Dashboards:** Super Admin (Owner), Front Desk Receptionist, Personal Trainer / Coach, Member Mobile Portal, and Public Marketing Site.
- **Modern Next.js / Tailwind Frontend:** High-converting landing page highlighting gym facilities, coach profiles, pricing packages, and online joining.

3. Commercial Quotation & Investment Options

We offer flexible investment models designed to maximize gym profitability while eliminating operational risks:

PACKAGE A: Managed Cloud Turnkey Model (Most Popular)

Low upfront capital expenditure with full cloud hosting, continuous updates, and hardware monitoring included.

Deliverable / Service Component	Scope & Specifications	Investment (LKR)
Turnkey Cloud Setup & Branding	Full cloud deployment, custom domain setup, database initialization, member roster import from Excel/CSV, 80mm thermal receipt branding.	LKR 75,000 *(One-time)*
Biometric Bridge & Turnstile Wiring	Local daemon deployment on Gym PC, SpeedFace face template synchronization, turnstile relay switch calibration, offline queue configuration.	LKR 45,000 *(One-time)*
FitMe Complete Cloud SaaS Subscription	Unlimited members, unlimited daily check-ins, Front Desk console, POS register, Sri Lankan EPF/ETF payroll engine, Sri Lankan nutrition database, daily cloud backups, 99.9% uptime SLA.	**LKR 18,500 / mo** ***(Billed Quarterly)***
Total Initial Launch Outlay	**Setup + Biometrics Integration + First Quarter Cloud Subscription (3 Months @ 18,500)**	**LKR 175,500**

PACKAGE B: Enterprise Perpetual License (Complete Ownership)

Designed for premium health clubs and fitness chains desiring self-hosted on-premises infrastructure and unlimited usage.

Deliverable / Service Component	Scope & Specifications	Investment (LKR)
Enterprise License & Source Deployment	Dedicated on-premises or private cloud deployment, full perpetual software license, multi-branch support capability, zero monthly fees.	**LKR 450,000** ***(One-time)***
Annual Maintenance & Support (Optional)	Priority 4-hour technical response, biometric driver maintenance, quarterly feature updates (waived in Year 1).	LKR 65,000 / yr *(Starting Year 2)*

4. Hardware Infrastructure Specifications

FitMe works with open, industry-standard equipment. Clients can utilize their existing equipment or request us to source commercial-grade units:

Hardware Component	Recommended Model / Minimum Specs	Estimated Cost (LKR)
Face ID / Biometric Terminal	ZKTeco SpeedFace-V5L [TI] (Dual-camera AI Face, Palm, RFID & Fingerprint), IP65 waterproof rated.	LKR 110,000 – 145,000 *(Direct vendor pricing)*
Optical Barrier Turnstile	Bridge Tripod or Flap Barrier Turnstile with integrated relay controller and bidirectional access.	LKR 280,000 – 420,000 *(Optional based on layout)*
Front Desk Reception PC	Intel Core i3/i5, 8GB RAM, 128GB SSD, Windows 10/11 Pro, Connected to Local Gym Network.	Existing Gym PC or LKR 65,000 (New)
Thermal POS Receipt Printer	80mm Thermal Receipt Printer (USB + Ethernet / ESC-POS standard, e.g., Xprinter / Epson).	LKR 18,000 – 24,000
Barcode Scanner	Handheld USB 1D/2D Barcode & QR Scanner with stand (for rapid retail checkout & stock audits).	LKR 8,500 – 12,000

5. Return on Investment (ROI) & Payback Model

Based on performance metrics gathered across active gyms with 250+ registered members, **FitMe pays for itself within 45 to 60 days** solely through automated leakage elimination:

Revenue Recovery Stream	Conservative Monthly Savings / Recovery	Annual Gym Impact
Eliminating Tailgating & Ghost Check-ins (15 overdue members paying LKR 5,000/mo)	**LKR 75,000 / month**	**LKR 900,000 / year**
POS Cash Drawer Reconciliation & Audit Control (Stops LKR 500 daily shrinkage)	**LKR 15,000 / month**	**LKR 180,000 / year**
Supplement & Beverage Margin Protection (Inventory cycle count enforcement)	**LKR 12,500 / month**	**LKR 150,000 / year**
HR & Payroll Calculation Labor Saved (25 managerial hours @ LKR 1,000/hr)	**LKR 25,000 / month**	**LKR 300,000 / year**
Total Monthly Estimated Net Recovery	**LKR 127,500 / month**	**LKR 1,530,000 / year**

6. 10-Day Deployment Roadmap & SLA

Deployment is executed smoothly without disrupting active gym hours:

Timeline	Stage	Key Milestones & Deliverables
Days 1 – 2	Cloud Provisioning	Cloud database spin-up, security configuration, custom domain mapping, membership plan data structuring.
Days 3 – 4	Data Migration	Import existing member records from Excel/CSV (active plans, balances, expiry dates, join dates).
Days 5 – 6	Hardware Bridge	Install ZKBio Bridge Daemon on Gym PC, calibrate SpeedFace terminal, wire turnstile relays, verify offline punch queue.
Days 7 – 8	POS & Thermal Printers	Setup 80mm ESC/POS thermal printers, configure retail product catalog, perform test checkout and cash audits.
Days 9 – 10	Staff Training & Go-Live	Front-desk receptionist & coach training, live member face enrollment, official turnstile gate activation.

7. Acceptance & Authorization Sign-off

By signing below, the Client authorizes the commencement of the FitMe Gym Operating System deployment under the terms outlined herein:

FOR THE CLIENT (GYM MANAGEMENT)	FOR FITME SYSTEM SOLUTIONS
Authorized Signature: _____	Authorized Signature: _____
Name: _____	Name: Sandaru Gangabada
Designation: _____	Designation: Lead Solutions Architect
Gym Name: _____	Company: FitMe Systems
Date: _____	Date: _____